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Tarek Hariri

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Education

Irvine, CA **University of California, Irvine** **September 2025 - Ongoing**
M.S. in Computer Science *GPA: 3.74*

William Howard Taft University **January 2025 - Ongoing**
Master of Business Administration

Irvine, CA **University of California, Irvine** **September 2020 - June 2024**
B.S. in Computer Science, Systems and Software Specialization *GPA: 3.7*

Awards

- Dean's Honor List for 10 quarters
- Regent's Scholar for four years

Research Experience

Graduate Researcher **Networking, Systems, and AI Lab (NetSAIL)** **Jan 2026 - Ongoing**

- Developing cluster-aware index maintenance scheduler for distributed ANN query search systems
- Models cluster relationships to allow nodes to leverage related nodes' resources, optimizing cluster resource utilization while protecting cluster from entering bad states
- Parallel orchestration layer built alongside Apache Solr, deployed on Kubernetes

Lab led by Professor Sangeetha Jyothi, https://ics.uci.edu/ sabdujyo/

Work Experience

Software Engineer, Intern **Rovren** **Oct 2023 - Feb 2024**

- Optimized software performance by implementing asynchronous callbacks into OpenCV source with **C++** and emscripten's Embind libraries.
- Enabled porting software app to web-app by configuring **Emscripten** toolchain to compile OpenCV to **WebAssembly**.
- Increased software performance by optimizing existing functions and developing new functionality into OpenCV source.
- Optimized OpenCV build configurations to minimize shipped application size

Projects

MCP-First LangGraph Agent for Entity Q&A | *Python, LangGraph*

- Implemented AI agent for entity-centric question answering over electronic health record data.
- Small, specialized LLM for intent classification and entity extraction.
- Performs on par with GPT-5.1 Thinking model with local computation– no cloud computing necessary.
- Layered architecture considerably improves determinism, auditability, and portability.

https://tarek-hariri.github.io/ophthalmology-agent/MCPLangGraph.pdf

GVAE and Transformer-Based Molecular Generation | *Python*

- Exploration of deep generative models for generation of drug-like, commercially available molecules.
- Preprocessing data with Quantitative Estimation of Drug-likeness (QED) prefixes for a causal Transformer model saw 100% valid, 100% unique, and 99.9% novel molecule generation with a Fréchet ChemNet Distance (FCD) score of 2.722.

https://tarek-hariri.github.io/molecular-generation/writeup.pdf

Fabflix | *HTML/CSS, jQuery, AJAX, Tomcat, AWS, EC2, SQL*
- Full-stack web application for browsing and searching movies
- Deployed on Tomcat running on an EC2 instance with MySQL integration for data storage
- Uses jQuery to call Java Servlet APIs for data retrieval following AJAX standards
- Implemented ReCaptcha and login cookies for user verification

https://github.com/tarek-hariri/fabflix-project

Crux Compiler | *Java, ANTLR*
- Wrote a grammar for Crux language and used ANTLR to generate parser
- Designed AST in **Java** that implements visitor pattern
- Constraint solving type checker
- High level intermediate representation generated using visitor pattern
- x86 assembly code generator

https://github.com/tarek-hariri/crux-compiler

Technical Skills

- **Languages:** C++, Python, SQL, C, Java, Javascript
- **Operating Systems:** Windows, Linux
- **Tools:** Git, AWS (EC2), Maven, Docker